CS316

INTRODUCTION TO AI AND DATA SCIENCE

Topic 8

K-fold Cross-validation and Grid Search

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Why one train/test split is not enough

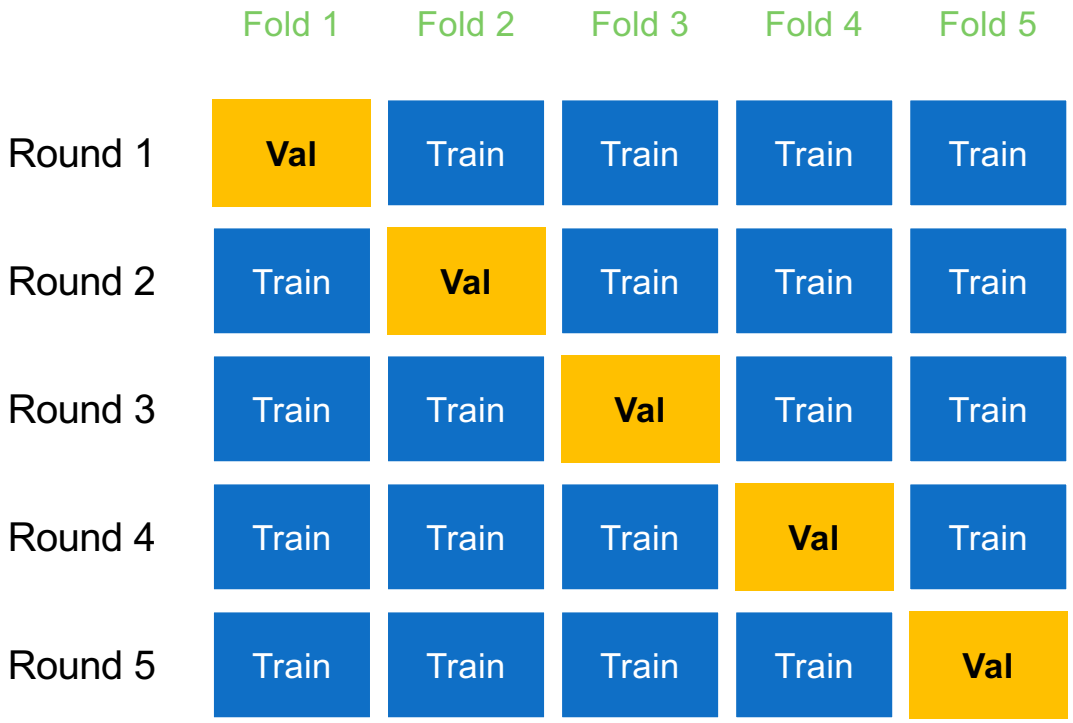
- A single split gives a single accuracy, and that number depends on which rows landed in the test set.
- Different random splits of the same data and model can give noticeably different scores (e.g. 0.95 to 0.99).
- Small datasets suffer most: one extra correct prediction can move accuracy by 1 to 2 points.
- If we also tune the model on the test set, we are no longer measuring the future; we are measuring our own choices.

The goal

Estimate how the model will perform on data it has never seen, and know how much that estimate can vary.

Cross-validation gives both a mean and a spread.

K-fold cross-validation: every sample validates once



Score = mean of the 5 validation scores, reported with their standard deviation.

How it works

- Split the training data into k equal folds.
- Hold out one fold for validation; train on the other k-1.
- Repeat k times so each fold is the validation set once.
- Average the k scores. The standard deviation tells you how stable the model is.

Grid search: trying every hyperparameter combination

- Parameters are learned from data (weights). Hyperparameters are set by us before training
- Grid search lists candidate values for each hyperparameter and tries every combination.
- Each combination is scored with k-fold CV, and the best mean score wins.
- Cost: combinations x k fits. Here 3 x 3 = 9 combinations x 5 folds = 45 fits.

param1 \ param2	**0.01**	**0.1**	**1**
1	0.91	0.93	0.88
10	0.94	**0.97**	0.92
100	0.93	0.95	0.90

Highlighted: best combination (C=10, gamma=0.1)